PHISHING EMAIL DETECTION & AWARENESS SYSTEM

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Course : Diploma in cyber security

Internship oraganization : Future Interns

Tools Used : Nmap , OWASAP ZAP , Browser DevTools

Date : April 2026

1. **INTRODUCTION**

Phishing is a type of cyber attack where attackers send fake emails pretending to be trusted organizations to steal sensitive information such as passwords, banking details, or personal data. These emails often create urgency to trick users.

2. **OBJECTIVES**

- Identify phishing indicators in emails
- Classify emails as Safe, Suspicious, or Phishing
- Understand phishing techniques
- Provide prevention and awareness guidelines

3. **TOOLS USED**

- Kali Linux
- WHOIS
- NSLOOKUP
- CURL
- NMAP
- Text Editor (Nano)

4. **METHODOLOGY**

- Created sample emails for analysis
- Extracted links from emails
- Analyzed domains using Kali Linux commands
- Identified suspicious patterns
- Classified emails based on risk

EMAIL 1 - SAFE

FIGURE 1

KEY FEATURES :

- Trusted sender (google.com)
- No suspicious link
- Normal message

CLASSIFICATION : SAFE

EMAIL 2 - SUSPICIOUS

FIGURE 2

FIGURE 3 - EXTRACT LINK

```
┌──(kali㉿kali)-[~]
└─$ grep -oP '(http|https)://[^\s]+' amazOn.txt
http://amazOn.com/login

┌──(kali㉿kali)-[~]
└─$
```

FIGURE 4 - WHOIS RESULT

```
┌──(kali㉿kali)-[~]
└─$ whois amaz0n.com
   Domain Name: AMAZON.COM
   Registry Domain ID: 281209_DOMAIN_COM-VRSN
   Registrar WHOIS Server: whois.markmonitor.com
   Registrar URL: http://www.markmonitor.com
   Updated Date: 2025-07-31T17:49:55Z
   Creation Date: 1994-11-01T05:00:00Z
   Registry Expiry Date: 2026-10-31T04:00:00Z
   Registrar: MarkMonitor Inc.
   Registrar IANA ID: 292
   Registrar Abuse Contact Email: abusecomplaints@markmonitor.com
   Registrar Abuse Contact Phone: +1.2086851750
   Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
   Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
   Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
   Domain Status: serverDeleteProhibited https://icann.org/epp#serverDeleteProhibited
   Domain Status: serverTransferProhibited https://icann.org/epp#serverTransferProhibited
   Domain Status: serverUpdateProhibited https://icann.org/epp#serverUpdateProhibited
   Name Server: NS1.AMZNDNS.CO.UK
   Name Server: NS1.AMZNDNS.COM
   Name Server: NS1.AMZNDNS.NET
   Name Server: NS1.AMZNDNS.ORG
   Name Server: NS2.AMZNDNS.CO.UK
   Name Server: NS2.AMZNDNS.COM
   Name Server: NS2.AMZNDNS.NET
   Name Server: NS2.AMZNDNS.ORG
   DNSSEC: unsigned
   URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-04-16T08:14:19Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar.  Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois
database through the use of electronic processes that are high-volume and
automated except as reasonably necessary to register domain names or
modify existing registrations; the Data in VeriSign Global Registry
Services' ("VeriSign") Whois database is provided by VeriSign for
information purposes only, and to assist persons in obtaining information
about or related to a domain name registration record. VeriSign does not
guarantee its accuracy. By submitting a Whois query, you agree to abide
by the following terms of use: You agree that you may use this Data only
```

FIGURE 5 - DNS LOOKUP RESULT

```
┌──(kali㉿kali)-[~]
└─$ nslookup amaz0n.com
Server:         192.168.1.1
Address:        192.168.1.1#53

Non-authoritative answer:
Name:   amaz0n.com
Address: 3.33.192.77
Name:   amaz0n.com
Address: 15.197.245.13
```

FIGURE 6 - HTTP RESPONSE

```
┌──(kali㉿kali)-[~]
└─$ curl -I http://amaz0n.com
HTTP/1.1 301 Moved Permanently
Server: awselb/2.0
Date: Thu, 16 Apr 2026 08:16:22 GMT
Content-Type: text/html
Content-Length: 134
Connection: keep-alive
Location: https://amaz0n.com:443/

┌──(kali㉿kali)-[~]
└─$
```

KEY FEATURES :

- Fake domain (amaz0n.com)
- Contains suspicious link
- Requests account verification

CLASSIFICATION : SUSPICIOUS

EMAIL 3 - PHISHING

FIGURE 7 - PHISHING EMAIL EXAMPLE

```
From: bank-alert@secure-login.net
Subject: Account Suspended

Dear Sir,

Your bank account has been suspended due to unusual activity.

Click here immediately to restore access:
http://secure-login.net

Failure to act now may result in permanent account closure.

Thank you
```

FIGURE 8 - DOMAIN INFORMATION

```
zsh: corrupt history file /home/kali/.zsh_history
┌──(kali㉿kali)-[~]
└─$ whois secure-login.net
   Domain Name: SECURE-LOGIN.NET
   Registry Domain ID: 2753920019_DOMAIN_NET-VRSN
   Registrar WHOIS Server: whois.cloudflare.com
   Registrar URL: http://www.cloudflare.com
   Updated Date: 2025-12-27T04:13:29Z
   Creation Date: 2023-01-26T15:28:39Z
   Registry Expiry Date: 2027-01-26T15:28:39Z
   Registrar: Cloudflare, Inc.
   Registrar IANA ID: 1910
   Registrar Abuse Contact Email: registrar-abuse@cloudflare.com
   Registrar Abuse Contact Phone: +1.6503198930
   Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
   Name Server: JARRED.NS.CLOUDFLARE.COM
   Name Server: KINSLEY.NS.CLOUDFLARE.COM
   DNSSEC: unsigned
   URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-04-16T08:17:35Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
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currently set to expire. This date does not necessarily reflect the expiration
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automated except as reasonably necessary to register domain names or
modify existing registrations; the Data in VeriSign Global Registry
Services' ("VeriSign") Whois database is provided by VeriSign for
information purposes only, and to assist persons in obtaining information
about or related to a domain name registration record. VeriSign does not
guarantee its accuracy. By submitting a Whois query, you agree to abide
by the following terms of use: You agree that you may use this Data only
for lawful purposes and that under no circumstances will you use this Data
to: (1) allow, enable, or otherwise support the transmission of mass
unsolicited, commercial advertising or solicitations via e-mail, telephone,
or facsimile; or (2) enable high volume, automated, electronic processes
that apply to VeriSign (or its computer systems). The compilation,
repackaging, dissemination or other use of this Data is expressly
prohibited without the prior written consent of VeriSign. You agree not to
use electronic processes that are automated and high-volume to access or
query the Whois database except as reasonably necessary to register
domain names or modify existing registrations. VeriSign reserves the right
to restrict your access to the Whois database in its sole discretion to ensure
```

FIGURE 9 - PORT SCAN RESULT

```
┌──(kali㉿kali)-[~]
└─$ nmap secure-login.net
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-16 04:19 -0400
Failed to resolve "secure-login.net".
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.74 seconds

┌──(kali㉿kali)-[~]
└─$
```

KEY FEATURES :

- Unknown sender
- Urgent and threatening message
- Contains malicious link
- Forces immediate action

5. PHISHING INDICATORS IDENTIFIED

- Fake email addresses
- Suspicious links
- Urgent messages
- Unknown domains

6. PHISHING TECHNIQUES

- **Email Spoofing:** Fake sender pretending to be trusted
- **Link Manipulation:** Fake links redirecting to malicious websites
- **Social Engineering:** Tricking users using emotions
- **Urgency Tactics:** Forcing quick decisions

7. RISK IMPACT

- Data theft
- Financial loss
- Identity theft
- Unauthorized access

8. PREVENTION & AWARENESS GUIDLINES

For Users:

- Verify sender email carefully
- Do not click unknown links
- Avoid downloading suspicious attachments
- Enable Two-Factor Authentication (2FA)

For Organizations:

- Provide cybersecurity awareness training
- Use email filtering systems
- Implement strong security policies

9. CONCLUSION

Phishing emails can be detected by analyzing sender details, links, and message content. Awareness and proper precautions can help prevent cyber attacks and protect user data.